UP 201 – Physics Laboratory

Experiment 5

Bragg's Law and Absorption of X-rays

Cu-anode X-ray unit • LiF and KBr single crystals • GM counter • Al absorbers

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28 Sept 2026

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1 Aim

1. To set up the X-ray unit, record the X-ray spectrum of the Cu anode with a LiF crystal, study the continuous and characteristic X-rays and determine the wavelength and energy of the characteristic K_α and K_β lines (with uncertainties).
2. To determine the inter-planar spacing of a KBr crystal using the characteristic lines.
3. To determine the attenuation of X-rays in aluminium and the mass attenuation coefficient μ/ρ as a function of wavelength.

2 Apparatus

PHYWE X-ray unit (35 kV maximum) with Cu-anode Coolidge tube, goniometer with 1:2 crystal–detector coupling, 1 mm and 5 mm apertures, LiF and KBr single crystals, Geiger–Müller (GM) counter tube, aluminium absorbers (0.02, 0.04, 0.06, 0.08 and 0.10 mm), computer with *measure* software.

3 Principle and Theory

3.1 Production of X-rays and the spectrum of an X-ray tube

Electrons emitted thermionically from the heated cathode are accelerated through the anode voltage U_A and strike the Cu anode with kinetic energy eU_A. Two kinds of radiation are produced.

Continuous spectrum (Bremsstrahlung). An electron decelerated in the Coulomb field of the anode nuclei radiates part of its energy as a photon; since the energy loss can take any value, the spectrum is continuous. The most energetic photon is emitted when the electron gives *all* its kinetic energy to a single photon (Duane–Hunt law):

$$eU_A = h\nu_{\max} = \frac{hc}{\lambda_{\min}} \quad \Longrightarrow \quad \lambda_{\min} = \frac{hc}{eU_A} = \frac{1.2398\ \text{nm kV}}{U_A}. \tag{1}$$

For $U_A = 35\,\text{kV}$ this gives $\lambda_{\min} = 35.42$ pm, *independent of the anode material.*

Characteristic spectrum. If an incident electron ejects a K-shell electron of a Cu atom, the vacancy is filled by an electron from a higher shell and the energy difference is emitted as a photon of fixed energy. For $L \to K$ and $M \to K$ transitions

$$E_{K_\alpha} = |E_K| - |E_L|, \qquad E_{K_\beta} = |E_K| - |E_M|, \tag{2}$$

which depend only on the atomic levels of Cu, not on U_A (provided eU_A exceeds the K-binding energy, $\approx 8.98\,\text{keV}$). They appear as sharp lines superimposed on the continuum. Literature values for Cu are $E_{K_\alpha} \approx 8.048\,\text{keV}$ ($\lambda = 154.2\,\text{pm}$) and $E_{K_\beta} \approx 8.905\,\text{keV}$ ($\lambda = 139.2\,\text{pm}$).

3.2 Crystal structure and Bragg's law

Derivation of Bragg's law. Consider a monochromatic beam of wavelength λ incident at glancing angle θ on a family of parallel lattice planes of spacing d (Fig. 1). Each plane reflects a small fraction of the beam specularly. Two rays reflected from adjacent planes travel different

distances: the lower ray covers the extra path $AB + BC$, where $AB = BC = d\sin\theta$. The path difference is therefore

$$\Delta = AB + BC = 2d\sin\theta. \tag{3}$$

The reflected rays reinforce each other (constructive interference) only if Δ is an integral number of wavelengths:

$$\boxed{2d\sin\theta = n\lambda}, \qquad n = 1, 2, 3, \ldots \tag{4}$$

Since the sum over many planes cancels the intensity at all other angles, sharp peaks appear at the Bragg angles $\theta_n = \arcsin\!\big(n\lambda/2d\big)$. A polychromatic beam is thus *dispersed*: each wavelength satisfies Eq. (4) at a different angle, which lets the crystal act as a spectrometer. Because $\sin\theta \le 1$, only wavelengths with $\lambda \le 2d/n$ are diffracted.

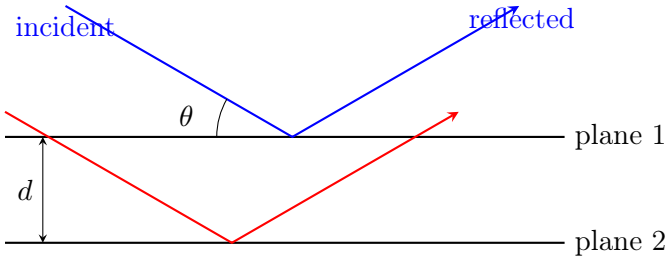


Figure 1: Geometry for Bragg's law (schematic, drawn as text: incident and reflected rays at glancing angle θ from planes separated by d; extra path $= 2d\sin\theta$).

Lattice spacing in a cubic crystal. For a cubic crystal of lattice constant a, the planes with Miller indices (hkl) have spacing

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}. \tag{5}$$

LiF, KBr and NaCl all have the rock-salt structure (two interpenetrating face-centred-cubic lattices of cations and anions). The strongest reflection is from the (200) planes, $d_{200} = a/2$; higher orders $n = 2, 3, 4$ of (200) are the (400), (600), (800) reflections. For LiF, $a \simeq 402.7\,\text{pm}$ so $d = 201.4\,\text{pm}$ (value used in the experiment).

1:2 **coupling.** In the goniometer the crystal is rotated through θ while the counter is rotated through 2θ, so the counter always sits at the direction of the reflected beam (angle between incident and reflected beams $= 2\theta$).

3.3 Geiger–Müller counter and dead-time correction

Each ionising photon produces a voltage pulse. After a pulse the GM tube is insensitive for a time τ (the *dead time*). Let N be the true rate of photons arriving and N_0 the measured rate. In one second the counter is dead for a total time $N_0\tau$ (each of the N_0 recorded pulses is followed by a dead interval τ), so it is live for $1 - N_0\tau$ seconds. Photons arriving during this live time are all counted:

$$N\,(1 - N_0\tau) = N_0 \quad \Longrightarrow \quad \boxed{N = \frac{N_0}{1 - \tau N_0}} \tag{6}$$

(inversely, $N_0 = N/(1+\tau N)$). The correction is negligible for $\tau N_0 \ll 1$ and diverges as $N_0 \to 1/\tau$. If the rate N_0 has statistical error $\sigma_{N_0} = \sqrt{N_0/t_g}$ for gate time t_g, then

$$\sigma_N = \frac{\partial N}{\partial N_0}\,\sigma_{N_0} = \frac{\sigma_{N_0}}{(1 - \tau N_0)^2}. \tag{7}$$

3.4 Absorption of X-rays

Beer–Lambert law. In a thin layer $\mathrm{d}x$ of an absorber the fractional loss of intensity is proportional to the thickness: $\mathrm{d}I = -\mu I\,\mathrm{d}x$, where μ is the linear attenuation coefficient. Integrating from $x = 0$ ($I = I_0$) to $x = d$:

$$\int_{I_0}^{I} \frac{\mathrm{d}I}{I} = -\mu \int_0^d \mathrm{d}x \implies I = I_0\,\mathrm{e}^{-\mu d} \quad \Longleftrightarrow \quad \ln I = \ln I_0 - \mu d. \tag{8}$$

A plot of $\ln I$ against d at fixed wavelength is a straight line of slope $-\mu$. Since absorption is proportional to the mass traversed, the density-independent *mass attenuation coefficient* μ/ρ ($\mathrm{cm^2\,g^{-1}}$) is used; $\rho_{\mathrm{Al}} = 2.70\,\mathrm{g\,cm^{-3}}$.

Interaction processes.

- *Photoelectric effect (PE):* the photon is absorbed by a bound electron, which is ejected with $E_{kin} = h\nu - E_{\mathrm{binding}}$.
- *Compton scattering (CS):* the photon scatters from a (quasi-free) electron and loses energy. Conservation of energy and momentum gives $\lambda' - \lambda = \dfrac{h}{m_0 c}\,(1 - \cos\varphi)$.
- *Pair production:* a photon creates an $e^- e^+$ pair in the field of a nucleus; energy conservation requires $h\nu \geq 2m_0c^2 = 1.022\,\mathrm{MeV}$.

In the keV range pair production is energetically forbidden, so

$$\mu = \mu_{PE} + \mu_{CS}. \tag{9}$$

For low-energy photons and a light absorber such as Al, PE dominates over CS. Between absorption edges the mass attenuation coefficient follows the empirical relation

$$\frac{\mu}{\rho} = C\,Z^3\lambda^3, \tag{10}$$

(Z = atomic number, C = constant); the λ^3 ($\propto E^{-3}$) dependence reflects the photoelectric cross-section. A plot of $\ln(\mu/\rho)$ versus $\ln\lambda$ should therefore be a straight line of slope ≈ 3.

3.5 Least-squares fitting and error propagation

For $y = a + bx$ with weights $w_i = 1/\sigma_i^2$:

$$b = \frac{S\,S_{xy} - S_x S_y}{S\,S_{xx} - S_x^2}, \quad a = \frac{S_y - bS_x}{S}, \quad \sigma_b = \sqrt{\frac{S}{S\,S_{xx} - S_x^2}}, \tag{11}$$

with $S = \sum w_i$, $S_x = \sum w_i x_i$, $S_y = \sum w_i y_i$, $S_{xx} = \sum w_i x_i^2$, $S_{xy} = \sum w_i x_i y_i$. When the reduced $\chi^2_\nu > 1$, σ_b is multiplied by $\sqrt{\chi^2_\nu}$. Other propagation formulae used (all derived from $\sigma_f^2 = \sum (\partial f/\partial x_i)^2 \sigma_i^2$):

$$\lambda = \frac{2d\sin\theta}{n} \qquad \Rightarrow \quad \sigma_\lambda = \lambda\cot\theta\,\sigma_\theta \quad (\sigma_\theta \text{ in rad}), \tag{12}$$

$$E = \frac{hc}{\lambda} = \frac{1239.84\ \mathrm{eV\,nm}}{\lambda} \qquad \Rightarrow \quad \frac{\sigma_E}{E} = \frac{\sigma_\lambda}{\lambda}, \tag{13}$$

$$\frac{\mu}{\rho} = \frac{\mu}{\rho} \qquad \Rightarrow \quad \sigma_{\mu/\rho} = \frac{\sigma_\mu}{\rho}. \tag{14}$$

4 Brief Procedure

1. The 1 mm aperture was mounted on the X-ray outlet, the LiF crystal placed in the goniometer holder and the goniometer calibrated (under the supervision of the instructor).
2. In the *measure* software: 1:2 coupling mode, gate time 2 s, angular step 0.1°, $U_A = 35\,\mathrm{kV}$, $I_A = 1\,\mathrm{mA}$; the spectrum was scanned from 6° to 55° (LiF) and the stored data corrected for dead time using Eq. (6).
3. The LiF crystal was replaced by KBr and scanned from 6° to 75°.
4. For absorption the aperture was changed to 5 mm and the LiF crystal was used. With each Al absorber (0, 0.02, 0.04, 0.06, 0.08, 0.10 mm) placed before the aperture, the spectrum was scanned from 15° to 25° (step 0.1°, gate 2 s) and stored.
5. The count rates at selected angles (wavelengths) were corrected for dead time and $\ln N$ vs. d was fitted to get μ and μ/ρ at each wavelength.

5 Observations

5.1 Experimental settings

Table 1: Settings used (read from the data files / measure project file).

	LiF spectrum	KBr spectrum	Al absorption
Anode voltage / current		35 kV / 1 mA	
GM tube voltage		500 V	
Gate time t_g		2 s	
Angular step		0.1°	
Scan range	6°–55°	6°–75°	15°–25°
Aperture	1 mm	1 mm	5 mm
Crystal	LiF	KBr	LiF
Dead time used		$\tau = 90\,\mu\mathrm{s}$ (see Sec. 8)	

Recorded quantity: count rate N_0 (counts s^{-1}) versus crystal angle θ. **Constants:** $d_{\mathrm{LiF}} = 201.4\,\mathrm{pm}$, $hc = 1239.84\,\mathrm{eV\,nm}$, $\rho_{\mathrm{Al}} = 2.70\,\mathrm{g\,cm}^{-3}$, $Z_{\mathrm{Al}} = 13$.

5.2 X-ray spectrum of Cu with LiF crystal

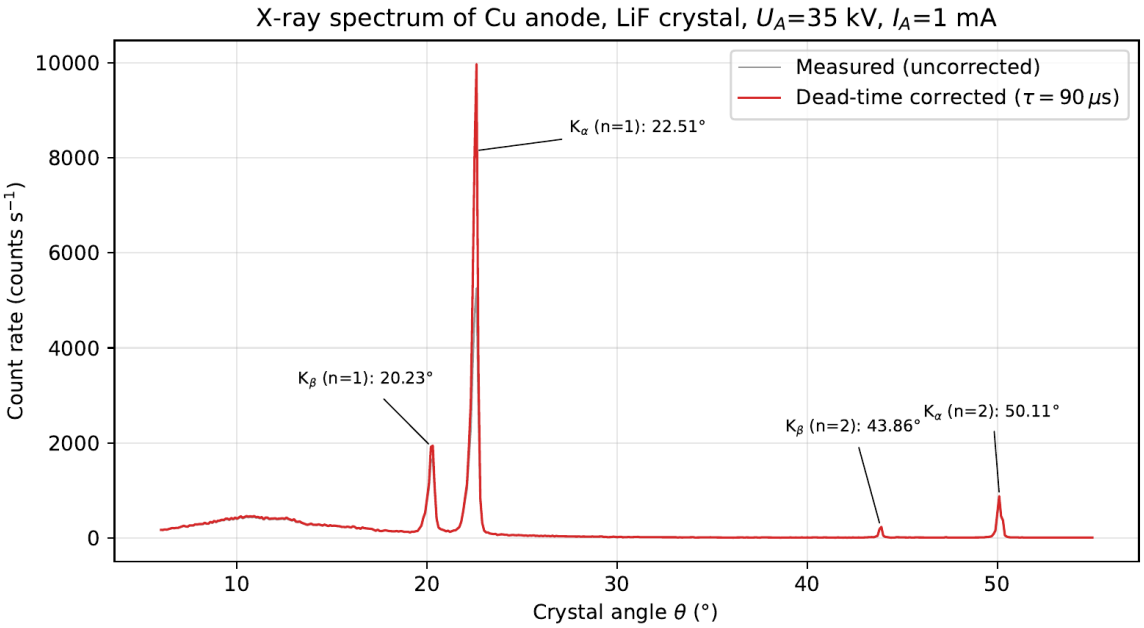


Figure 2: X-ray spectrum of the Cu anode from the LiF crystal: measured (grey) and dead-time corrected (red). The continuum maximum is near 10°; K_β and K_α lines appear in first ($n = 1$) and second ($n = 2$) order.

The positions of the four characteristic peaks were obtained by fitting a Gaussian on a linear background to the dead-time corrected data around each peak (Fig. 3).

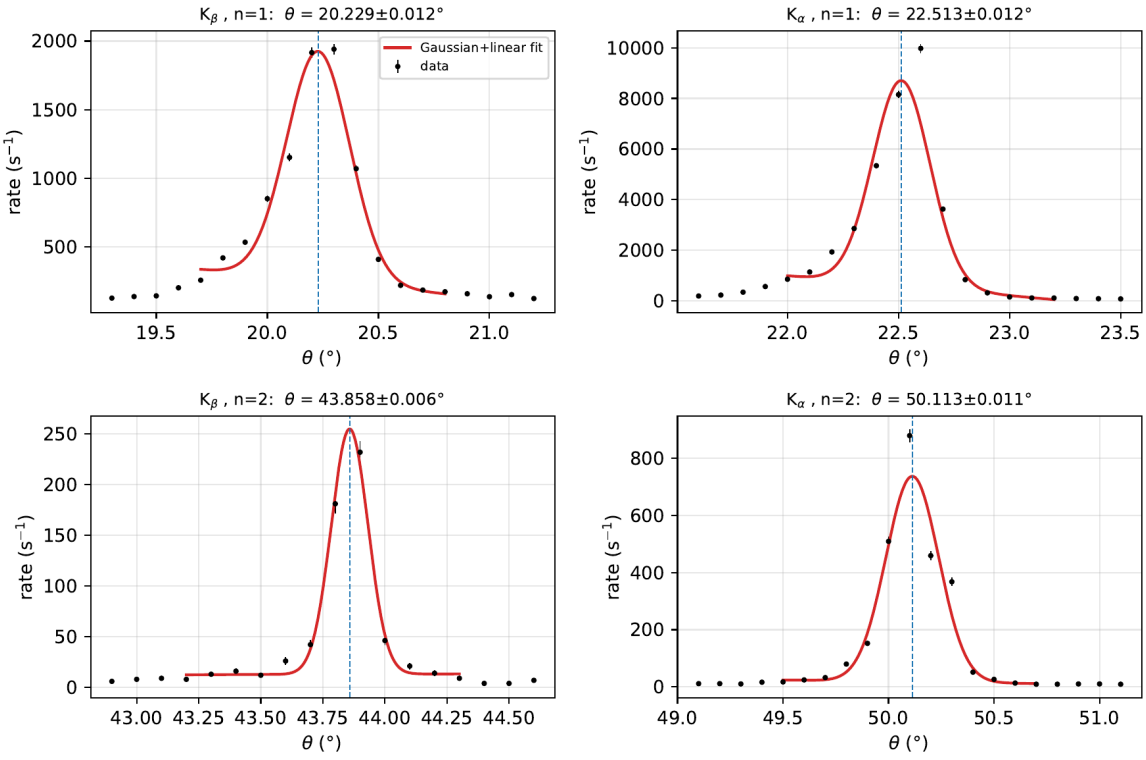


Figure 3: Gaussian-plus-linear fits to the K_β and K_α peaks of LiF in first and second order (dead-time corrected data with Poisson error bars).

Table 2: Characteristic peaks from the LiF crystal. σ_θ(fit) is the fit uncertainty; the total $\sigma_\theta = \sqrt{\sigma_{\text{fit}}^2 + (0.05°)^2}$ includes half the angular step. FWHM and peak rate refer to the corrected data (peak rate = fitted peak height including background).

Line	n	θ (°)	σ_θ (fit)	σ_θ (total)	FWHM (°)	Rate (s^{-1})	λ (pm)	E (keV)
K_β	1	20.229	0.012	0.052	0.33	1925	139.28 ± 0.34	8.902 ± 0.022
K_α	1	22.513	0.012	0.051	0.30	8703	154.23 ± 0.33	8.039 ± 0.017
K_β	2	43.858	0.006	0.050	0.17	255	139.54 ± 0.13	8.885 ± 0.008
K_α	2	50.113	0.011	0.051	0.29	737	154.54 ± 0.12	8.023 ± 0.006

5.3 X-ray spectrum with KBr crystal

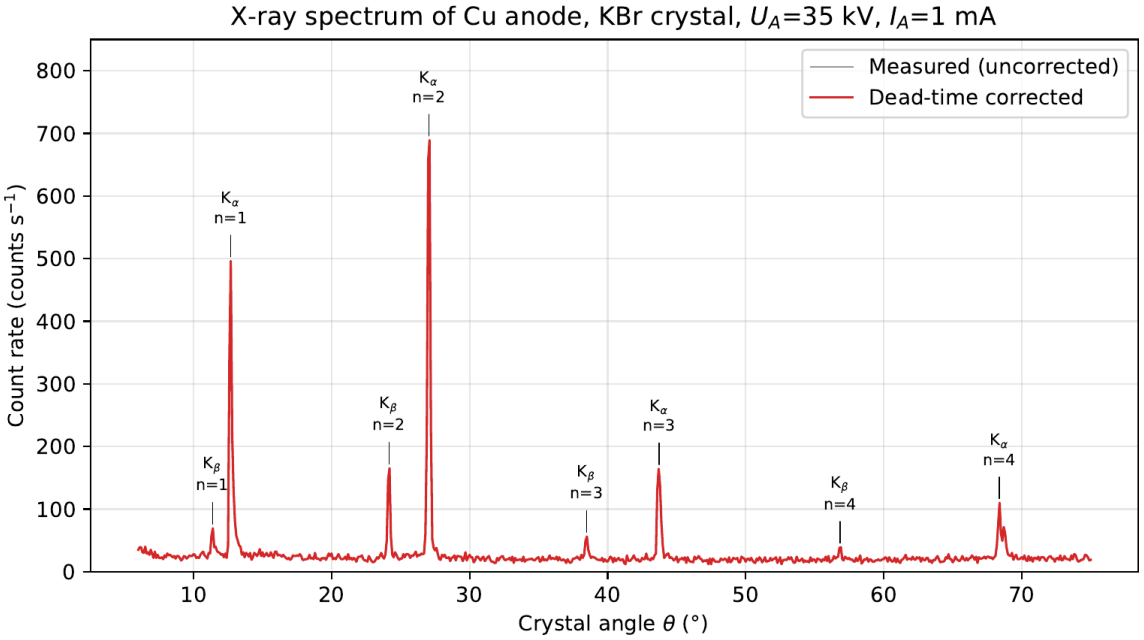


Figure 4: X-ray spectrum of the Cu anode from the KBr crystal (measured and dead-time corrected). Eight peaks are seen: K_β and K_α in orders $n = 1$ to 4.

Table 3: Peak positions for KBr. The wavelengths are the mean LiF values from Sec. 6.2. $d_{\text{naive}} = n\lambda/(2\sin\theta)$ assumes zero angular offset. The last column is the residual $\theta_{\text{obs}} - \theta_{\text{fit}}$ of the final fit (Sec. 6.3). The K_β ($n = 4$) peak is weak (one-bin peak), so $\sigma_\theta = 0.10°$ was assigned.

Line	n	λ (pm)	θ_{obs} (°)	σ_θ (°)	$\sin\theta_{\text{obs}}$	d_{naive} (pm)	Residual (°)
K_β	1	139.41	11.370	0.052	0.1971	353.6	-0.003
K_α	1	154.38	12.691	0.051	0.2197	351.4	-0.014
K_β	2	139.41	24.156	0.050	0.4092	340.7	-0.000
K_α	2	154.38	27.051	0.050	0.4548	339.5	-0.006
K_β	3	139.41	38.472	0.052	0.6221	336.1	-0.001
K_α	3	154.38	43.729	0.051	0.6912	335.0	+0.020
K_β	4	139.41	56.851	0.100	0.8372	333.0	+0.073
K_α	4	154.38	68.380	0.056	0.9297	332.1	-0.019

5.4 Absorption of X-rays in aluminium

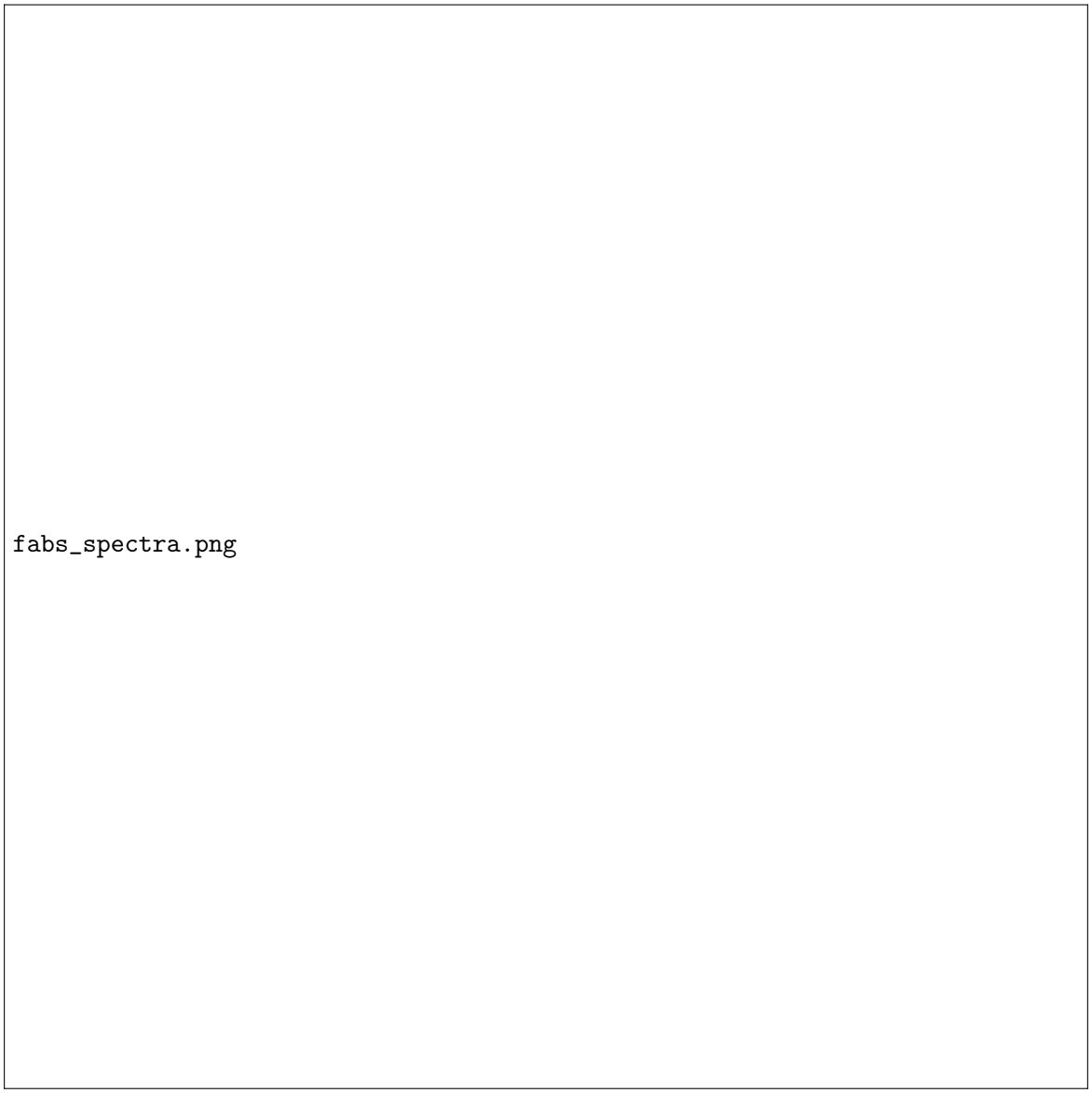


Figure 5: Dead-time corrected spectra between 15° and 25° for the six absorber thicknesses. The intensity falls monotonically with thickness at every angle.

Data used for the attenuation analysis. Ten angles (wavelengths) were selected. For continuum angles the rate is the mean of five neighbouring points ($\theta \pm 0.2°$, i.e. 10 s of counting) to reduce Poisson noise; the K_β point (20.2°) is a single point because the peak is steep. The 22°–23.2° region (K_α peak) was *not* used: raw rates there reach $1.4 \times 10^4\,\mathrm{s}^{-1}$, which exceeds $1/\tau$ for $\tau = 90\,\mu\mathrm{s}$, so Eq. (6) cannot be applied (Sec. 8). The file named `lif_4mm` was taken as the 0.04 mm absorber (its rates lie between those of 0.02 and 0.06 mm at every angle).

Table 4: Measured (uncorrected) count rate N_0 (s^{-1}) at the selected angles for different Al thickness d.

	d (mm)					
θ (°)	0	0.02	0.04	0.06	0.08	0.10
15.0	841.7	847.3	754.7	724.0	660.3	646.3
16.0	773.6	763.8	696.2	653.0	580.4	574.6
17.0	687.8	667.4	599.6	553.4	503.2	482.8
18.0	645.0	620.2	560.8	505.0	460.0	431.8
19.0	623.2	549.4	494.2	449.2	400.6	390.0
20.2	2938.0	2358.0	2020.0	1769.0	1434.0	1294.0
21.0	851.6	742.0	626.2	552.8	462.0	439.4
23.5	772.2	629.8	527.2	447.4	359.6	334.4
24.0	504.6	420.8	343.6	303.0	263.8	255.0
25.0	412.3	337.3	277.3	253.0	211.3	204.7

Table 5: Dead-time corrected rate $N = N_0/(1 - \tau N_0)$ (s^{-1}), $\tau = 90\,\mu\mathrm{s}$.

	d (mm)					
θ (°)	0	0.02	0.04	0.06	0.08	0.10
15.0	910.6	917.3	809.7	774.5	702.1	686.3
16.0	831.5	820.2	742.7	693.8	612.4	605.9
17.0	733.2	710.0	633.8	582.4	527.1	504.7
18.0	684.7	656.9	590.6	529.0	479.9	449.3
19.0	660.2	578.0	517.2	468.1	415.6	404.2
20.2	3994.1	2993.2	2468.8	2104.0	1646.5	1464.6
21.0	922.3	795.1	663.6	581.7	482.0	457.5
23.5	829.9	667.6	553.5	466.2	371.6	344.8
24.0	528.6	437.4	354.6	311.5	270.2	261.0
25.0	428.2	347.9	284.4	258.9	215.4	208.5

6 Calculations

6.1 Dead-time correction (sample)

For the LiF spectrum at $\theta = 22.5°$ (on the K_α ($n = 1$) peak) the measured rate was $N_0 = 4701\ \mathrm{s}^{-1}$. With $\tau = 90\,\mu\mathrm{s}$, $\tau N_0 = 0.423$ and from Eq. (6)

$$N = \frac{4701}{1 - 90 \times 10^{-6} \times 4701} = 8149\ \mathrm{s}^{-1} \quad \text{(factor 1.73)}.$$

The correction was applied to every point of every spectrum, which is why the peaks in Figs. 2–4 are taller in the corrected curves.

6.2 Wavelength and energy of the characteristic lines (LiF)

Using Eq. (4) with $d = 201.4\,\mathrm{pm}$, for K_α ($n = 1$, $\theta = 22.513°$):

$$\lambda = \frac{2d \sin\theta}{n} = 2(201.4)\sin(22.513°) = 154.23\ \mathrm{pm}, \qquad \sigma_\lambda = \lambda \cot\theta\, \sigma_\theta = 0.33\ \mathrm{pm},$$

$$E = \frac{hc}{\lambda} = \frac{1239.84\ \text{eV nm}}{0.154.23\ \text{nm}} \approx 8.039\ \text{keV} \quad (\sigma_E = 0.017\ \text{keV}).$$

The same calculation for the other peaks is listed in Table 2. The first- and second-order results for each line agree within their uncertainties, so the mean (with the larger of the standard error and the combined statistical error) is quoted:

$$\lambda_{K_\alpha} = 154.38 \pm 0.15\ \text{pm}, \qquad E_{K_\alpha} = 8.031 \pm 0.008\ \text{keV},$$
$$\lambda_{K_\beta} = 139.41 \pm 0.13\ \text{pm}, \qquad E_{K_\beta} = 8.893 \pm 0.009\ \text{keV}.$$

Continuum limit. By Eq. (1), $\lambda_{\min} = 35.42\,\text{pm}$ for 35 kV, i.e. the first-order short-wavelength cutoff lies at $\theta_{\min} = \arcsin(\lambda_{\min}/2d) = 5.05°$, *below* the start of the scan (6°); the cutoff therefore could not be located in this data set.

6.3 Inter-planar spacing of KBr

Eight peaks (K_α and K_β, $n = 1$–4) are observed. Using the LiF wavelengths, the spacing from each peak is $d = n\lambda/(2\sin\theta)$ (Table 3). These values are *not* constant: they fall from $\approx 353\,\text{pm}$ ($n = 1$) to $\approx 332\,\text{pm}$ ($n = 4$). A constant spacing combined with a constant angular offset δ of the crystal zero explains this trend, because an offset matters most at small θ. The data were therefore fitted with

$$\theta_{\text{obs}} = \arcsin\left(\frac{n\lambda}{2d}\right) + \delta \tag{15}$$

with d and δ free (weighted non-linear least squares; σ_θ from Table 3). Result (Fig. 6):

$$d = 330.3\ \text{pm}\ (\sigma_{\text{fit}} = 0.15), \qquad \delta = -0.81° \pm 0.03°, \qquad \chi^2_\nu = 0.15.$$

For comparison, forcing $\delta = 0$ gives $d = 333.6\,\text{pm}$ but with $\chi^2_\nu = 111$, i.e. an unacceptable fit. The wavelength uncertainty (common to all peaks) shifts d by $\pm 0.33\,\text{pm}$, so

$$\boxed{d_{\text{KBr}} = 330.3 \pm 0.4\ \text{pm}} \qquad [(200)\ \text{planes};\ a = 2d].$$

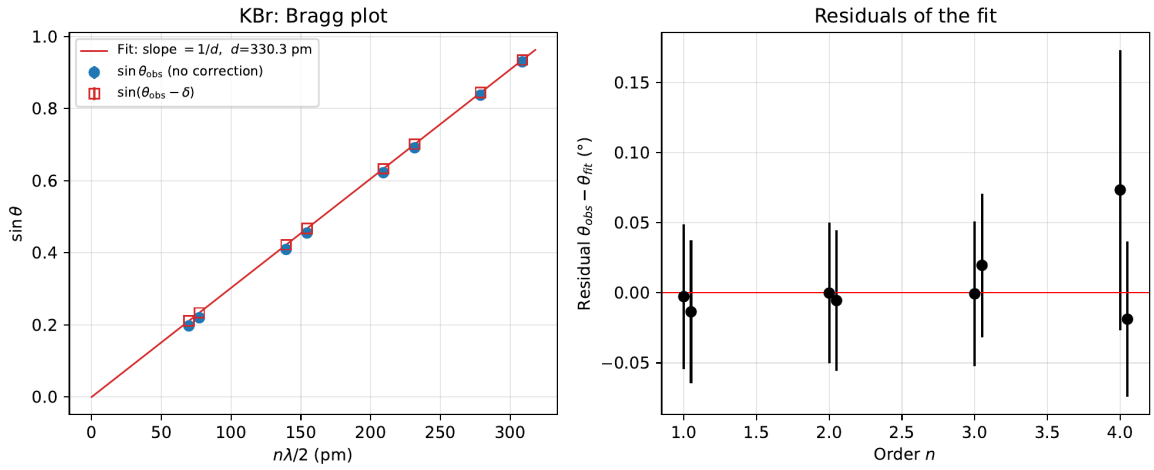


Figure 6: Left: Bragg plot for KBr. $\sin\theta$ against $n\lambda/2$ is a straight line of slope $1/d$ once the angular offset δ is removed. Right: residuals of the fit.

6.4 Attenuation coefficient and mass attenuation coefficient of Al

At each selected angle θ the wavelength follows from Eq. (4) ($n = 1$, $d = 201.4\,\text{pm}$), $\ln N$ was plotted against d (Fig. 7), and the weighted straight-line fit gave $-\mu$ [Eq. (8)]. Weights come

from Eq. (7), $\sigma_{\ln N} = \sigma_N/N$. Example, $\theta = 16^\circ$ ($\lambda = 111.0\,\mathrm{pm}$): slope $\mu = 36.0\ \mathrm{cm}^{-1}$, so $\mu/\rho = 36.0/2.70 = 13.3\ \mathrm{cm}^2\,\mathrm{g}^{-1}$ (see Table 6).

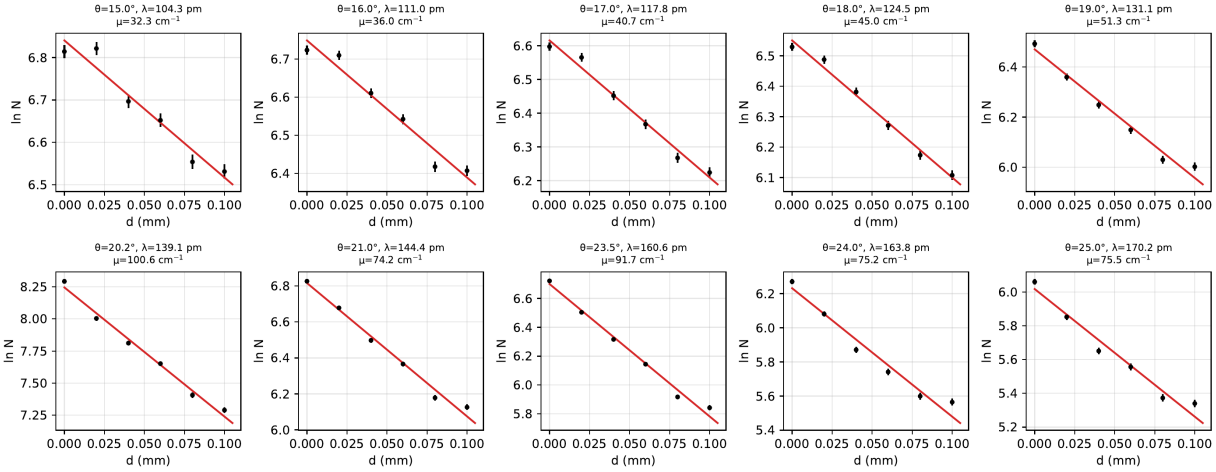


Figure 7: $\ln N$ against absorber thickness d for the ten selected angles; the slope of each weighted straight-line fit is $-\mu$.

Table 6: Attenuation coefficients of aluminium. Method A: weighted fit of $\ln N$ vs. d (Eq. 8). The reference column is an indicative value from tabulated NIST data (see text). Method B: fit of $N = S\mathrm{e}^{-\mu d} + B$ with a constant background B, shown only where the fit was well behaved. χ^2_ν refers to Method A; μ/ρ in $\mathrm{cm}^2\,\mathrm{g}^{-1}$.

θ (°)	λ (pm)	E (keV)	μ (cm^{-1})	μ/ρ (A)	χ^2_ν	Ref. μ/ρ	μ/ρ (B)
15.0	104.3 ± 0.3	11.89	32.3 ± 3.9	12.0 ± 1.4	4.1	15.8	–
16.0	111.0 ± 0.3	11.17	36.0 ± 3.8	13.3 ± 1.4	5.9	19.0	–
17.0	117.8 ± 0.3	10.53	40.7 ± 2.8	15.1 ± 1.0	2.8	22.5	–
18.0	124.5 ± 0.3	9.96	45.0 ± 2.6	16.7 ± 1.0	2.3	26.5	–
19.0	131.1 ± 0.3	9.45	51.3 ± 3.6	19.0 ± 1.3	3.9	30.9	49.3 ± 10.9
20.2	139.1 ± 0.3	8.91	100.6 ± 5.6	37.3 ± 2.1	5.6	36.7	–
21.0	144.4 ± 0.3	8.59	74.2 ± 4.0	27.5 ± 1.5	5.9	40.9	45.1 ± 12.2
23.5	160.6 ± 0.3	7.72	91.7 ± 4.4	34.0 ± 1.6	6.2	55.9	52.5 ± 9.7
24.0	163.8 ± 0.3	7.57	75.2 ± 6.9	27.9 ± 2.5	10.7	59.2	72.0 ± 10.3
25.0	170.2 ± 0.3	7.28	75.5 ± 6.8	28.0 ± 2.5	8.5	66.2	70.2 ± 13.1

Dependence on wavelength. Taking logarithms of Eq. (10), $\ln(\mu/\rho) = \ln(CZ^3) + n\ln\lambda$ with expected $n = 3$.

Table 7: Data for the μ/ρ versus λ graphs (Method A).

θ (°)	λ (nm)	λ^3 (10^{-3} nm^3)	μ/ρ	$\sigma_{\mu/\rho}$	$\ln\lambda$	$\ln(\mu/\rho)$
15.0	0.1043	1.133	11.97	1.44	-2.261	2.482
16.0	0.1110	1.369	13.32	1.41	-2.198	2.590
17.0	0.1178	1.633	15.06	1.03	-2.139	2.712
18.0	0.1245	1.928	16.68	0.96	-2.084	2.814
19.0	0.1311	2.255	19.01	1.32	-2.031	2.945
20.2	0.1391	2.691	37.28	2.08	-1.973	3.618
21.0	0.1444	3.008	27.49	1.47	-1.936	3.314
23.5	0.1606	4.144	33.95	1.64	-1.829	3.525
24.0	0.1638	4.398	27.85	2.54	-1.809	3.327
25.0	0.1702	4.933	27.97	2.50	-1.771	3.331

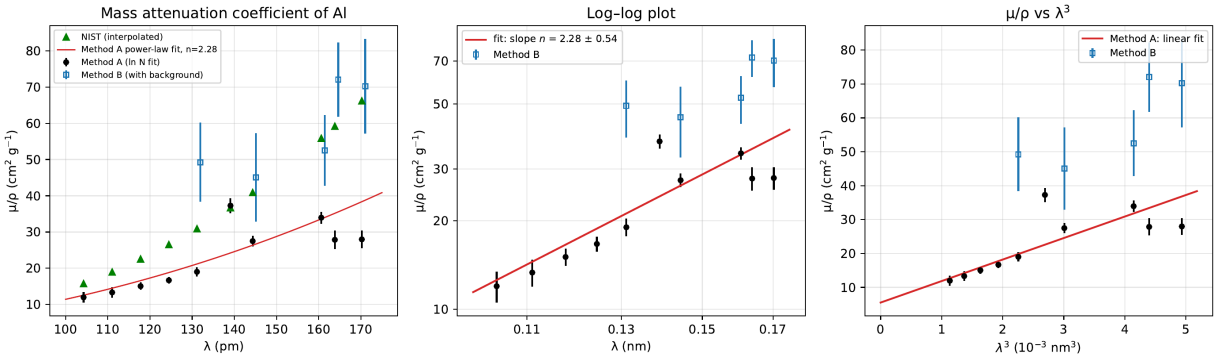


Figure 8: Dependence of the mass attenuation coefficient of Al on wavelength: linear plot (left), log–log plot (centre) and μ/ρ against λ^3 (right). Black circles: Method A; blue squares: Method B; green triangles: indicative reference values.

The weighted log–log fit gives

$$n = 2.28 \pm 0.54, \qquad \frac{\mu}{\rho} = 2161\,\lambda^{2.28} \ (\lambda \text{ in nm},\ \mu/\rho \text{ in cm}^2\,\text{g}^{-1}),$$

which differs from the expected value 3 by 1.3σ. A straight-line fit of μ/ρ against λ^3 (right panel) gives a slope of (6344 ± 1380) cm^2 g^{-1} nm^{-3} and an intercept of 5.5 ± 3.4 cm^2 g^{-1} (consistent with zero within about 2σ). Using Eq. (10) with $Z = 13$ ($Z^3 = 2197$), slope $= CZ^3$ gives $C = (2.89 \pm 0.63)$ cm^2 g^{-1} nm^{-3}.

7 Results

Table 8: Summary of results. Literature values for Cu: $E_{K_\alpha} = 8.048\,\text{keV}$, $E_{K_\beta} = 8.905\,\text{keV}$; KBr $a = 660.0\,\text{pm}$ ($d_{200} = 330.0\,\text{pm}$).

Quantity	Measured	Literature (deviation)
λ_{K_α}	154.38 ± 0.15 pm	154.2 pm
E_{K_α}	8.031 ± 0.008 keV	8.048 keV (-0.21%)
λ_{K_β}	139.41 ± 0.13 pm	139.2 pm
E_{K_β}	8.893 ± 0.009 keV	8.905 keV (-0.13%)
$\lambda_{\min}$ (35 kV, from Eq. 1)	35.42 pm (not within scan)	–
d_{KBr}	330.3 ± 0.4 pm	330.0 pm (0.08%)
Angular offset of KBr crystal δ	$-0.81^\circ \pm 0.03^\circ$	–
Exponent n in $\mu/\rho \propto \lambda^n$	2.28 ± 0.54	3
μ/ρ at K_β ($\lambda \approx 139\,\text{pm}$)	$37.3 \pm 2.1\ \text{cm}^2\,\text{g}^{-1}$	$\approx 36.7\ \text{cm}^2\,\text{g}^{-1}$

8 Error Analysis and Discussion

Characteristic lines (LiF). The angular uncertainty has two parts: the peak-centre uncertainty from the fit ($\approx 0.005^\circ$–0.013°) and the finite resolution of the goniometer (step 0.1°, taken as $\pm 0.05^\circ$). The latter dominates, giving $\sigma_\lambda = \lambda \cot\theta\, \sigma_\theta$ of 0.1–0.3 pm, i.e. about 0.1% to 0.2% of λ (larger at small θ since $\cot\theta$ is larger; second-order peaks are more precise). The measured E_{K_α} is -0.21% and E_{K_β} is -0.13% away from the literature values. The K_α line is in fact an unresolved doublet ($K_{\alpha 1}$, $K_{\alpha 2}$ at 154.06 and 154.44 pm), so the fitted value is an intensity-weighted mean and sits within the doublet.

Dead time. The write-up gives Eq. (6) but no value of τ; $\tau = 90\,\mu\text{s}$ (typical quoted value for this type of GM tube) was assumed. Two remarks: (i) the effect on the results is small, since the correction changes the *height* of the peaks but hardly their *position*: the fitted K_α ($n = 1$) angle is 22.518°, 22.515°, 22.513° and 22.511° for $\tau = 0$, 60, 90 and 120 µs – a total change of 0.008°, much smaller than $\sigma_\theta = 0.05^\circ$; (ii) the raw rates in the absorption scans near the K_α line (N_0 up to $1.4 \times 10^4\,\text{s}^{-1}$) are larger than $1/\tau = 1.1 \times 10^4\,\text{s}^{-1}$ for $\tau = 90\,\mu\text{s}$, which is impossible for a true non-paralysable dead time. This implies that the actual τ of the counter must be below $\approx 70\,\mu\text{s}$, or that the counter is not purely non-paralysable at these rates. The peak-height information near K_α in the absorption scans was therefore not used. For the continuum points used in the attenuation analysis $\tau N_0 \le 0.08$ (correction $\le 9\%$), and for the K_β point $\tau N_0 = 0.26$ (correction 36%); since the correction is applied to every thickness, its influence on the *slope* μ is much smaller than on the individual rates. If the instructor supplies a different τ, only the peak heights and these small corrections change.

KBr spacing. The statistical uncertainty of the fit (with $\sigma_\theta = 0.05^\circ$ assigned to each peak, which overestimates the true scatter since $\chi^2_\nu \ll 1$) is 0.15 pm. The uncertainty in the LiF-derived wavelengths contributes 0.33 pm, giving a total of ± 0.4 pm, i.e. $\approx 0.1\%$. The result agrees with the literature value 330.0 pm (0.08%). The naive single-peak values (332–354 pm, up to 7% too high) are biased by the angular offset $\delta = -0.81^\circ$; this is a systematic error produced by the seating/orientation of the KBr crystal relative to the goniometer zero, and the fit of Fig. 6 removes it. An offset δ changes $\sin\theta$ by $\cos\theta\,\delta$, a relative change $\delta \cot\theta$ that is large for the $n = 1$ peaks and small at large θ; using several orders is therefore what allows d and δ to be separated.

Attenuation coefficient. The statistical (Poisson) error per rate is $\sigma_{N_0} = \sqrt{N_0/t_g}$ (about 2–3% for 10 s of effective counting at $N_0 \approx 700\,\mathrm{s}^{-1}$), and it was propagated with Eq. (7) to $\sigma_{\ln N}$ and then to σ_μ by the weighted fit. For many angles $\chi^2_\nu > 1$ (Table 6), showing that there are errors beyond counting statistics; the quoted σ_μ were multiplied by $\sqrt{\chi^2_\nu}$. The wavelength uncertainty $\sigma_\lambda \approx 0.3\,\mathrm{pm}$ (0.05° in θ) is negligible on the λ scale of Fig. 8.

The measured μ/ρ increases with λ (Fig. 8), as expected, but the exponent $n = 2.28 \pm 0.54$ is lower than the expected 3 and the continuum points ($\lambda > 120\,\mathrm{pm}$) lie below the indicative literature values, while the K_β point (20.2°, where the signal is strong) agrees (37.3 ± 2.1 against $\approx 36.7\,\mathrm{cm}^2\,\mathrm{g}^{-1}$). The reason is probably a non-ideal "background" of radiation that does not obey $\lambda = 2d\sin\theta$ with $n = 1$ and is only weakly absorbed by the foil. This makes the apparent attenuation smaller than the true one. Likely contributions are:

- *Higher-order reflection:* at angle θ the crystal also reflects $\lambda/2$, $\lambda/3$, … (the tube at 35 kV emits down to 35 pm). This harder radiation is far less absorbed ($\mu/\rho \propto \lambda^3$), so it increases the transmitted fraction.
- *Scattered/fluorescent radiation* from the crystal, air and housing, which is detected at all angles. A background term fit (Method B, $N = S\mathrm{e}^{-\mu d} + B$) gives $B \approx 165$–$305\,\mathrm{s}^{-1}$ at $\theta \geq 19°$, and then μ/ρ at 24° and 25° becomes $\approx 70\,\mathrm{cm}^2\,\mathrm{g}^{-1}$ (versus ≈ 28 with Method A), close to the indicative reference (≈ 60–66), but with large errors (Table 6); at $\theta < 19°$ the background term is unconstrained by the data.
- *Other:* the large 5 mm aperture (poor angular resolution), non-uniform foil thickness and the nominal thicknesses (tolerance not given; foil thickness errors of $\sim 5\%$ would give an equal relative error in μ), and the GM counter's energy-dependent efficiency.

Because of these systematic effects, the main quantitative absorption result should be regarded as the μ/ρ value at the K_β line and the qualitative λ^n dependence ($n \approx 2.3 \pm 0.5$, consistent with 3 at about 1.3σ), not as a precise determination of C.

Reference values. The "reference" μ/ρ values are taken from the NIST X-ray mass attenuation tables for Al (50.3, 26.2 and $7.96\,\mathrm{cm}^2\,\mathrm{g}^{-1}$ at 8, 10 and 15 keV), interpolated linearly in log–log (extrapolated below 8 keV with the local slope), and should be treated as indicative only.

9 Conclusion

- The LiF spectrum shows a continuum with sharp Cu K_β and K_α lines in first and second order. The measured energies $E_{K_\alpha} = 8.031 \pm 0.008\,\mathrm{keV}$ and $E_{K_\beta} = 8.893 \pm 0.009\,\mathrm{keV}$ agree with the literature values to within −0.21% and −0.13%.
- The KBr spectrum has eight peaks (orders 1–4 of K_α and K_β); a fit that includes a crystal angular offset of −0.81° gives $d_{200} = 330.3 \pm 0.4\,\mathrm{pm}$, in agreement with the literature value of 330.0 pm.
- The intensity of the X-ray beam falls exponentially with Al thickness (Beer–Lambert law) and μ/ρ increases with wavelength with an exponent $n = 2.28 \pm 0.54$ (expected 3). The K_β line gives $\mu/\rho = 37.3 \pm 2.1\,\mathrm{cm}^2\,\mathrm{g}^{-1}$ in good agreement with the tabulated value; the weaker continuum points are biased low by higher-order and scattered radiation.